🔬 B.Sc. MICROBIOLOGY – 3rd SEMESTER

📘 FINAL EXAM PREPARATION NOTES

🧫 UNIT I: GENERAL MICROBIOLOGY

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1. Historical Introduction

Explanation:
Microbiology developed from early ideas that diseases were caused by supernatural forces. Later, scientists proved that **microorganisms cause diseases**. The invention of the microscope led to the discovery of bacteria and other microbes.

Examples:

- Leeuwenhoek observed bacteria
- Pasteur proved fermentation
- Koch identified disease-causing bacteria

Keywords:
History of microbiology, spontaneous generation, germ theory, pioneers, microscopes, Koch's postulates

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2. Microscopy and Morphology of Bacteria

Explanation:
Microscopy deals with instruments used to observe bacteria which are invisible to naked eyes. Morphology describes **shape, size, structure, and arrangement** of bacteria.

Examples:

- Shapes: cocci (round), bacilli (rod)
- Arrangements: chains, clusters

Keywords:
Light microscope, electron microscope, staining, cocci, bacilli, spirilla, flagella, capsule

3. Growth and Nutrition of Bacteria

Explanation:
Bacterial growth occurs by **binary fission**. Growth depends on nutrients like carbon, nitrogen, minerals, water, and energy source.

Examples:

- Growth curve: lag → log → stationary → death
- Autotrophs use CO_2; heterotrophs use organic carbon

Keywords:
Binary fission, growth curve, nutrients, autotrophs, heterotrophs, enzymes

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4. Culture Media and Culture Methods

Explanation:
Culture media provide nutrients for bacterial growth in laboratories. Culture methods help isolate and identify bacteria.

Examples:

- Nutrient agar
- MacConkey agar
- Blood agar

Keywords:
Culture media, selective, differential, enrichment, anaerobic culture

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5. Processing of Specimens

Explanation:
Clinical specimens must be properly collected, transported, and processed to detect pathogens accurately.

Examples:

- Blood for septicemia
- Urine for UTI

Keywords:
Aseptic collection, transport media, contamination, centrifugation

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6. Microbial Pathogenicity

Explanation:
Pathogenicity is the ability of microorganisms to cause disease. It depends on virulence factors.

Examples:

- Toxins
- Enzymes
- Capsules

Keywords:
Pathogenicity, virulence, toxins, adhesion, invasion, host response

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🧬 UNIT II: IMMUNOLOGY

(Already expanded but rewritten simpler)

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7. Immunity

Explanation:
Immunity is the body's defense mechanism against infections.

Examples:

- Innate immunity: skin
- Acquired immunity: antibodies

Keywords:
Innate immunity, acquired immunity, active, passive

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8. Antigen

Explanation:
Antigens are substances that stimulate immune response when they enter the body.

Examples:
Bacterial proteins, viral antigens

Keywords:
Antigenicity, epitope, haptens

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9. Antibodies (Immunoglobulins)

Explanation:
Antibodies are proteins produced by plasma cells to neutralize antigens.

Examples:
IgG, IgA, IgM

Keywords:
Immunoglobulins, heavy chain, light chain

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10. Antigen–Antibody Reactions

Explanation:
Specific interaction between antigen and antibody used in diagnosis.

Examples:
Agglutination, precipitation

Keywords:
Specificity, affinity, ELISA

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11. Complement System

Explanation:
A group of serum proteins that assist antibodies in destroying pathogens.

Examples:
Cell lysis, inflammation

Keywords:
Classical pathway, alternative pathway, MAC

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12. Structure and Functions of Immune System

Explanation:
Immune system consists of organs, tissues, and cells responsible for immunity.

Examples:
Thymus, spleen

Keywords:
Primary organs, secondary organs, lymphocytes

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13. Immune Response

Explanation:
Response of immune system to foreign antigens.

Examples:
Primary and secondary response

Keywords:
Memory cells, humoral immunity

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14. Hypersensitivity

Explanation:
Exaggerated immune reactions causing tissue damage.

Examples:
Allergy, TB reaction

Keywords:
Type I–IV hypersensitivity

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15. Autoimmunity

Explanation:
Body produces antibodies against self-antigens.

Examples:
Rheumatoid arthritis

Keywords:
Autoantibodies, self-tolerance

🦠 UNIT III: SYSTEMIC BACTERIOLOGY

(VERY IMPORTANT FOR FINAL EXAM)

For ALL bacteria write in exam as:
Morphology → Culture → Pathogenicity → Lab diagnosis → Prevention

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16. Staphylococcus

Explanation:
Gram-positive cocci arranged in clusters. Causes pus-forming infections.

Examples:
Boils, abscess

Keywords:
Gram-positive cocci, clusters, coagulase, toxins

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17. Streptococcus and Enterococcus

Explanation:
Gram-positive cocci in chains. Cause throat and heart infections.

Examples:
Sore throat, endocarditis

Keywords:
Chains, hemolysis, Lancefield classification

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18. Pneumococcus

Explanation:
Encapsulated Gram-positive diplococci causing pneumonia.

Keywords:
Diplococci, capsule, optochin sensitive

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19. Neisseria

Explanation:
Gram-negative diplococci causing meningitis and gonorrhea.

Keywords:
Oxidase positive, intracellular

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20. Corynebacterium

Explanation:
Club-shaped bacteria causing diphtheria.

Keywords:
Metachromatic granules, toxin

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21. Bacillus

Explanation:
Large spore-forming aerobic bacteria.

Keywords:
Spores, anthrax

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22. Clostridium

Explanation:
Anaerobic spore-forming bacteria producing powerful toxins.

Keywords:
Anaerobic, tetanus, botulinum toxin

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23. Non-sporing Anaerobes

Explanation:
Anaerobic bacteria without spores, part of normal flora.

Keywords:
Bacteroides, anaerobic infections

24. Enterobacteriaceae

Explanation:
Gram-negative rods found in intestine.

Examples:
E. coli, Salmonella

Keywords:
Lactose fermentation, endotoxin

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25. Vibrio

Explanation:
Comma-shaped bacteria causing cholera.

Keywords:
Alkaline media, toxin

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26. Pseudomonas

Explanation:
Opportunistic pathogen causing hospital infections.

Keywords:
Pigment, non-fermenter

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27. Yersinia

Explanation:
Causes plague.

Keywords:
Bipolar staining

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28. Haemophilus

Explanation:
Fastidious bacteria requiring X and V factors.

Keywords:
Chocolate agar

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29. Bordetella

Explanation:
Causes whooping cough.

Keywords:
Pertussis toxin

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30. Brucella

Explanation:
Zoonotic bacteria causing undulant fever.

Keywords:
Intracellular, zoonosis

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31. Mycobacterium

Explanation:
Acid-fast bacteria causing tuberculosis.

Keywords:
Mycolic acid, Ziehl-Neelsen stain

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32. Spirochaetes

Explanation:
Thin spiral bacteria causing syphilis.

Keywords:
Motility, Treponema

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33. Mycoplasma and Ureaplasma

Explanation:
Smallest bacteria lacking cell wall.

Keywords:
Pleomorphic, no cell wall

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34. Actinomycetes

Explanation:
Filamentous bacteria resembling fungi.

Keywords:
Sulfur granules

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35. Miscellaneous Bacteria

Explanation:
Includes bacteria not fitting classical groups.

Keywords:
Atypical bacteria

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36. Rickettsia, Orientia and Coxiella

Explanation:
Obligate intracellular bacteria transmitted by arthropods.

Keywords:
Typhus, intracellular

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37. Chlamydia and Chlamydophila

Explanation:
Intracellular bacteria with unique life cycle.

Keywords:
Elementary body, reticulate body

🧪 UNIT IV: VIROLOGY

38. General Properties of Viruses

Explanation:
Viruses are non-cellular infectious agents that require host cells.

Keywords:
Capsid, envelope, obligate intracellular

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39. DNA Viruses

Explanation:
Viruses containing DNA genome.

Keywords:
Herpesvirus, nuclear replication

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40. RNA Viruses

Explanation:
Viruses with RNA genome.

Keywords:
Influenza, cytoplasmic replication

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41. Hepatitis Viruses

Explanation:
Viruses causing liver inflammation.

Keywords:
Jaundice, Hepatitis A–E

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42. Retroviruses – HIV

Explanation:
RNA virus using reverse transcriptase enzyme.

Keywords:
CD4 cells, AIDS

🍄 UNIT V: MYCOLOGY

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43. Medical Mycology

Explanation:
Study of fungi causing diseases in humans.

Examples:
Candidiasis, ringworm

Keywords:
Yeasts, molds, dimorphic fungi, mycoses

UNIT VI: PARASITOLOGY

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44. Medical Parasitology

Explanation

Medical parasitology is the branch of microbiology that deals with **parasites causing diseases in humans**. A parasite is an organism that lives **in or on another organism (host)** and depends on the host for food and survival, causing harm.

Parasites are classified into:

1. **Protozoa** – single-celled organisms
2. **Helminths** – multicellular worms
3. **Arthropods** – insects acting as parasites or vectors

Parasites have **life cycles**, often involving **more than one host**, which is important for disease transmission and control.

Examples

- *Entamoeba histolytica* → Amoebic dysentery
- *Plasmodium* → Malaria
- *Ascaris lumbricoides* → Roundworm infection
- Mosquito → Vector for malaria

Keywords

Parasite, host, definitive host, intermediate host, protozoa, helminths, ectoparasite, endoparasite, life cycle, vector

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🏥 UNIT VII: HOSPITAL INFECTION CONTROL AND SAFETY

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45. Healthcare Associated Infection (HAI)

Explanation

Healthcare Associated Infections (HAIs) are infections that patients acquire **during hospital stay or medical treatment** and were **not present at the time of admission**. These infections occur due to contaminated equipment, poor hygiene, or invasive procedures.

HAIs increase **hospital stay, cost, and mortality** and are a major public health problem.

Examples

- Urinary tract infection due to catheter
- Pneumonia in ventilated patients
- Surgical site infections

Keywords

Nosocomial infection, hospital-acquired infection, cross infection, invasive procedures, pathogens

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46. Standard Precautions

Explanation

Standard precautions are **basic infection prevention measures** applied to **all patients**, irrespective of their diagnosis. These precautions assume that **blood and body fluids may be infectious**.

They protect both **patients and healthcare workers**.

Examples

- Wearing gloves while handling blood
- Safe disposal of sharps
- Use of masks and gowns

Keywords

Universal precautions, PPE, hand hygiene, blood and body fluids, infection prevention

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47. Hand Hygiene

Explanation

Hand hygiene refers to cleaning hands using soap and water or alcohol-based hand rubs to remove microorganisms. It is the **single most effective method** to prevent HAIs.

Poor hand hygiene is the **main cause of infection spread** in hospitals.

Examples

- Washing hands before and after patient contact
- Using alcohol rubs in ICU

Keywords

Hand washing, alcohol-based hand rub, WHO five moments, antiseptic

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48. Transmission-Based Precautions

Explanation

Transmission-based precautions are **additional infection control measures** used when infections spread through specific routes.

Types

1. **Contact precautions** – skin contact
2. **Droplet precautions** – cough/sneeze
3. **Airborne precautions** – aerosols

Examples

- TB → Airborne
- Influenza → Droplet
- MRSA → Contact

Keywords

Contact transmission, droplet transmission, airborne infection, isolation

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49. Major Healthcare Associated Infection Types

Explanation

These are the **most common infections** seen in hospitals due to medical devices and procedures.

Examples

- CAUTI – Catheter-associated UTI
- CLABSI – Central line bloodstream infection
- SSI – Surgical site infection
- VAP – Ventilator-associated pneumonia

Keywords

CAUTI, CLABSI, SSI, VAP, invasive devices

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50. HAI Surveillance (Including HICC)

Explanation

Surveillance is the **systematic collection and analysis of infection data** to control HAIs. HICC (Hospital Infection Control Committee) monitors infection trends and implements control policies.

Functions of HICC

- Infection monitoring
- Training staff
- Antibiotic policy

Keywords

Surveillance, infection rates, HICC, monitoring, reporting

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51. Sterilisation and Disinfection

Explanation

- **Sterilisation** destroys **all forms of life**, including spores
- **Disinfection** destroys **most pathogens**, but not spores

These processes are essential to prevent infection transmission.

Examples

- Autoclave → Sterilisation
- Phenol → Disinfection

Keywords

Sterilisation, disinfection, autoclave, antiseptic, spores

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52. Biomedical Waste Management

Explanation

Biomedical waste includes waste generated during diagnosis, treatment, or immunization. Improper disposal can cause **infection and environmental pollution**.

Waste is segregated using **color-coded bags**.

Examples

- Yellow bag → Infectious waste
- Red bag → Plastic waste

Keywords

Biomedical waste, segregation, color coding, disposal, incineration

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53. Needle Stick Injury

Explanation

Needle stick injury is accidental injury caused by needles contaminated with blood. It poses risk of **blood-borne infections**.

Examples

- Injury during injection
- Recapping needles

Keywords

Sharps injury, HIV, Hepatitis B, post-exposure prophylaxis (PEP)

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54. Environmental Surveillance

Explanation

Environmental surveillance monitors **hospital surroundings** like air, water, and surfaces to identify infection sources.

Examples

- Air sampling in OT
- Surface swabs

Keywords

Environmental monitoring, contamination, infection source

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55. Laundry Management

Explanation

Laundry management involves safe handling, washing, and disinfection of hospital linen to prevent infection spread.

Keywords

Soiled linen, washing, disinfection, infection control

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56. Immunisation of Healthcare Workers

Explanation

Healthcare workers are at high risk of infections. Vaccination protects them and prevents disease transmission to patients.

Examples

- Hepatitis B vaccine
- Influenza vaccine

Keywords

Immunisation, occupational safety, vaccines, prevention

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57. Antimicrobial Stewardship

Explanation

Antimicrobial stewardship ensures **rational and responsible use of antibiotics** to reduce antimicrobial resistance.

Keywords

Rational antibiotic use, resistance prevention, antibiotic policy

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58. Patient Safety

Explanation

Patient safety focuses on preventing **medical errors, infections, and harm** during healthcare delivery.

Keywords

Quality care, risk reduction, patient protection

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59. Safety Protocols for Healthcare Workers

Explanation

These are guidelines to protect healthcare workers from **biological, chemical, and physical hazards**.

Keywords

PPE, biosafety, occupational health, safety guidelines

🧪 UNIT VIII: APPLIED MICROBIOLOGY

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60. Normal Microbial Flora of the Human Body

Explanation

Normal flora are microorganisms that **normally live on the skin and mucous membranes** without causing disease. They protect against pathogens and help maintain health.

Examples

- *E. coli* in intestine
- *Staphylococcus epidermidis* on skin

Keywords

Normal flora, commensals, opportunistic pathogens, colonization

61. Blood Stream Infections

Explanation

Bloodstream infections occur when microorganisms enter the blood, leading to systemic illness.

Examples

- Septicemia
- Bacteremia

Keywords

Sepsis, blood culture, bacteremia

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62. Infective Syndromes: Laboratory Diagnosis

Explanation

Diagnosis is based on **clinical symptoms and laboratory tests** rather than identifying a single organism.

Examples

- Fever → blood culture
- Diarrhea → stool examination

Keywords

Syndromic approach, clinical diagnosis, laboratory tests

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63. Specimen Collection and Transport

Explanation

Proper specimen collection and transport are essential for accurate diagnosis. Poor handling leads to false results.

Keywords

Aseptic technique, transport media, contamination prevention

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64. Prophylactic Immunisation

Explanation

Vaccination given to **prevent diseases before exposure**.

Examples

- BCG → TB
- Polio vaccine

Keywords

Vaccination, immunisation schedule, disease prevention

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65. Antibiotic Resistance and Antimicrobial Sensitivity Testing

Explanation

Antibiotic resistance occurs when bacteria survive antibiotic action. AST helps select the correct antibiotic.

Examples

- Kirby–Bauer disc diffusion test

Keywords

Antibiotic resistance, AST, MIC, disc diffusion

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66. Vehicles and Vectors

Explanation

- **Vehicles** transmit pathogens indirectly

- **Vectors** transmit pathogens directly

Examples

- Water → vehicle
- Mosquito → vector

Keywords

Transmission, vectors, vehicles, disease spread

📘 UNIT IX: MICROBIOLOGY AT A GLANCE

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67. Essentials of Microbiology at a Glance

Explanation

This unit provides a **quick overview of important microbiology concepts** for revision before exams.

Keywords

Microorganisms, infection, immunity, diagnosis

🧪 UNIT X: PRACTICAL MICROBIOLOGY

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68. Practical Microbiology

Explanation

Practical microbiology involves laboratory techniques used for **identification, cultivation, and study of microorganisms**.

Examples

- Gram staining
- Culture techniques
- Biochemical tests

Keywords

Staining, microscopy, culture methods, laboratory diagnosis